

Tide retrospective: the cubic-order remainder layer (gamma-rung programme, stage 2a)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Stage 2 of the gamma-rung programme upgrades the seabed's rescaled integral asymptotics from first to second order: quadratising $e^{-s} \approx 1 - s + s^2/2$ instead of linearising improves the integrated remainder from $O(1/t)$ to $O(1/(t\sqrt{t}))$, which is what lets the A^2 coefficient at order $1/t$ become visible in the J_n expansions. The stage split at its natural seam during execution: this tide (2a) builds the remainder layer; the next (2b) assembles the decomposition and the second-order J_n headline.

2 The thing formalised

Six declarations in `Laplace/OneD/IntegralRemainder2.lean`, sorry-free with zero warnings, mirroring the first-order remainder layer declaration by declaration. The pointwise bound¹ feeds the signed exponential remainder of stage 1 at order three; the cube of the rescaled perturbation is dominated at scale $1/(t\sqrt{t})$ by even powers²; the combined bound³ routes through the existing two-branch Gaussian decay (`rescaled_max_decay`, reused verbatim); and the integrated form⁴ gives

$$\left| \int u^n e^{-u^2/2} (e^{-s_t(u)} - (1 - s_t(u) + s_t(u)^2/2)) du \right| \leq \frac{K}{t\sqrt{t}} \quad (t \geq 1),$$

the master estimate that stage 2b will consume. The error scale is kept `rpow`-free as $K/(t\sqrt{t})$, consistent with the seabed's $1/\sqrt{t}$ style.

3 Proof strategy

Layer by layer as in the first-order file: Taylor bound times Gaussian weight, then the decay lemma replaces $e^{-u^2/2} \max(1, e^{-s})$ by $e^{-c_0 u^2}$, then the cube bound $(x+y)^3 \leq 4(x^3+y^3)$ with $(\sqrt{t})^3 = t\sqrt{t}$ on the cubic branch and $t^3 \geq t\sqrt{t}$ on the quartic one, and finally the integral bound by domination against the integrable even-power family. The monotonicity cascade runs on `gcongr` throughout.

¹`perturbation_remainder3.pointwise`.

²`rescaled_cube_bound`, with `abs_pow_nine.le` absorbing $|u|^9 \leq u^8 + u^{10}$.

³`perturbation_remainder3.combined`.

⁴`perturbation_remainder3.integral_bound`.

4 Roadblocks and resolutions

Six build iterations, the programme’s heaviest so far, all at the elaboration level: `abs_add` is now `abs_add_le`; `gcongr` discharges its side conditions from the context, so trailing `exacts` die; `positivity` needs even powers in $(x^k)^2$ shape and an atom’s positivity in context before it will sign a cube; a conclusion-only implicit parameter needed explicit passing (the recurring anchor lesson); and one edit misfired into a stray untracked `Laplace.lean` at the repository root (junk from an earlier session, now deleted) rather than the worktree — caught immediately by the failed build, with the canonical clone verified clean.

5 What was Mathlib, what was new

Mathlib: `gcongr`, the integrable Gaussian-power family, integral monotonicity. Seabed: the entire architecture — the first-order layer supplied the proof skeleton, and `rescaled_max_decay` the analytic content. Stage 1 supplied the pointwise input. New: the even-power absorption and the bookkeeping.

6 Lessons

`gcongr` is the right engine for monotonicity cascades over products and quotients of nonnegative quantities: five of the six lemmas’ inequality steps reduced to it once stated in factored form. When it can see the hypothesis it needs, it uses it — do not follow it with an `exact`.

7 Follow-ups

- Stage 2b: the quadratised six-term decomposition (the $\sqrt{t}$ -atom identity for s_t^2) and the second-order J_n headline consuming this tide’s integral bound.
- Stages 3–6 per the programme plan.